Paper 2 - Assessing Realistic Issues for Product Development

Content

In this assignment, you will assess issues related to product development and commercialization for your senior design project. You will explore professional, ethical, legal, security, social, global, economic, and other issues important in transitioning your project from a prototype to a product.

As part of your write up for the issue you choose, you can consider its relationship to public health, safety, and welfare, as well as global, cultural, social, environmental, or economic factors.

Below are some sample issues grouped two categories: **Security** and **Commercialization**. Each team member must write about a unique issue, and your team must discuss at least 2 security issues and at least 2 commercialization issues (e.g. you could have 2 members pick security issues and 4 pick commercialization issues).

Security Issues

- **Payment** In e-commerce, there would be a need to protect personally identifiable information (PII) including names, addresses, accounts, credit card numbers, etc. If your project involves selling a product, you must consider payment issues and security by reviewing available commercial products (e.g., for encryption or secure payment) or other security techniques (e.g., Smart Card - see http://www.smartcardalliance.org). This should include, in the case of products, the cost associated with using a product or service (e.g., Paypal).
- **HIPAA Security**: In health care, the Health Insurance Portability and Accountability Act (HIPAA) provides a set of security guidelines in the usage, transmission, and sharing of protected health information (PHI). If your project involves individuals that may have to enter personal information (social security, income data, health care data, etc.), you may have to deal with HIPAA regulations (see https://www.hhs.gov/hipaa and http://www.hipaa.org/). For this issue, you need to research the impact and implications of such issues for your project.
- **Privacy and Confidentiality**: This issue differs from HIPAA security issues since it is involving the privacy and confidentiality of personal, financial, and health information. Data privacy involves a determination an understanding of what information can be shared between various parties. Confidentiality refers to the techniques necessary to protect the privacy of information. How do you determine who is allowed to see what information? How do you balance strict information protection via encryption with availability?
- **Authorization and Authentication**: Authorization refers to who is allowed to utilize what information at what times. Authentication involves the process that verifies the identity of user in which information they are authorized to utilize. Who is allowed to authorize access to information? The application owner? User? What level of authentication is supported? Two factor? Multi factor?

Commercialization Issues

- **Funding**: If your final project is a prototype, how would go about producing commercial-grade product for your project? Will you work on the commercial release gratis? Will you seek external funding? If so, who would you approach; e.g., will seek a bank loan, a (federal) grant, a (military-derived) contract?
- **Commercialization**: How would you go about the process of commercializing your project? Would you do so on your own, e.g., by forming a company? Approach a company or venture capitalist? Release an

open source or executable evaluation copy? What considerations are their in terms of ownership? How would you price your product? What features would your product need to be commercially viable? Can you estimate the man-hours needed to transition from a prototype to a product? Would you do phase releases? These and many other questions must be addressed.

- **Intellectual Property (IP)**: Who will own the IP rights resulting from your project? Will you attempt to patent any discoveries or processes? How will you make sure that you do not infringe on existing patents? Did you use university resources? If your instructor provided comments that were then incorporated into your project, would that be a university resource? What arrangements will need to be made with the university if you used university resources?
- **Ethical/Legal**: Many projects often involve users providing their login identity (user name and password), with the application taking that information and logging on on their behalf to various web services and/or databases. For example, a project that involves a university student with one login being able to access multiple systems, or a project for electronic payment that allows one individual to set up a single password to be used to pay utilities, credit cards, mortgages, etc., all on-line. For this issue, you must consider the ethical and legal issues for your project in this regard.
- **Software Licensing**: There are many different models that are emerging in regards to software licensing, including per user basis, site license, subscription (web and non-web), hosting of data, open source, etc. One critical software license in the area is the humanitarian free and open source software. Will you sell the rights to your system to another firm and receive royalties, or will you market your system directly? How will open-source software be handled and released?

Length

4 pages (~1200 words) excluding works cited.

Revisions

You will do peer-review of one of your teammate's documentation with the rubric below. You will then revise your documentation, at which point I will grade it using the same rubric.

Note that the grade you get during peer-review is not factored into your grade - there is no incentive to give higher or lower marks to your teammates. There is incentive to giving accurate marks - it will help you become a better writer, and your teammates will be able to get a better grade if they have good feedback.

Rubric

- Content (50%)
  - Explores one Security or Commercialization issue relevant to your project.
- Style (20%)
  - Content is aimed at a management level audience (technically proficient but may not have a Bachelor's in Computer Science)
- Citations (20%)
  - Paper contains at least 5 appropriate citations. Citations are properly and consistently sited and included in a works cited section at the end of the paper.
- Formatting (10%)
  - Paper is 4 pages, double-spaced with 1 inch margins.
  - Formatting is consistent and professional throughout.